

PID Control in Robotics

PID Purpose

A PID controller regulates a system using the error between a desired setpoint and measured output.

Proportional Term

The proportional term produces a response based on the current error.

Integral Term

The integral term accumulates past error and can reduce steady-state error.

Derivative Term

The derivative term responds to the rate of change of error and can reduce overshoot.

Applications

PID control is commonly used for motor speed, robot position, orientation, robotic arms, and balancing robots.